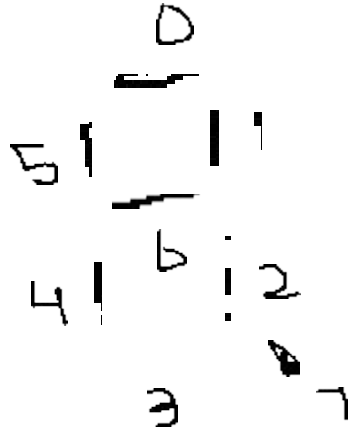


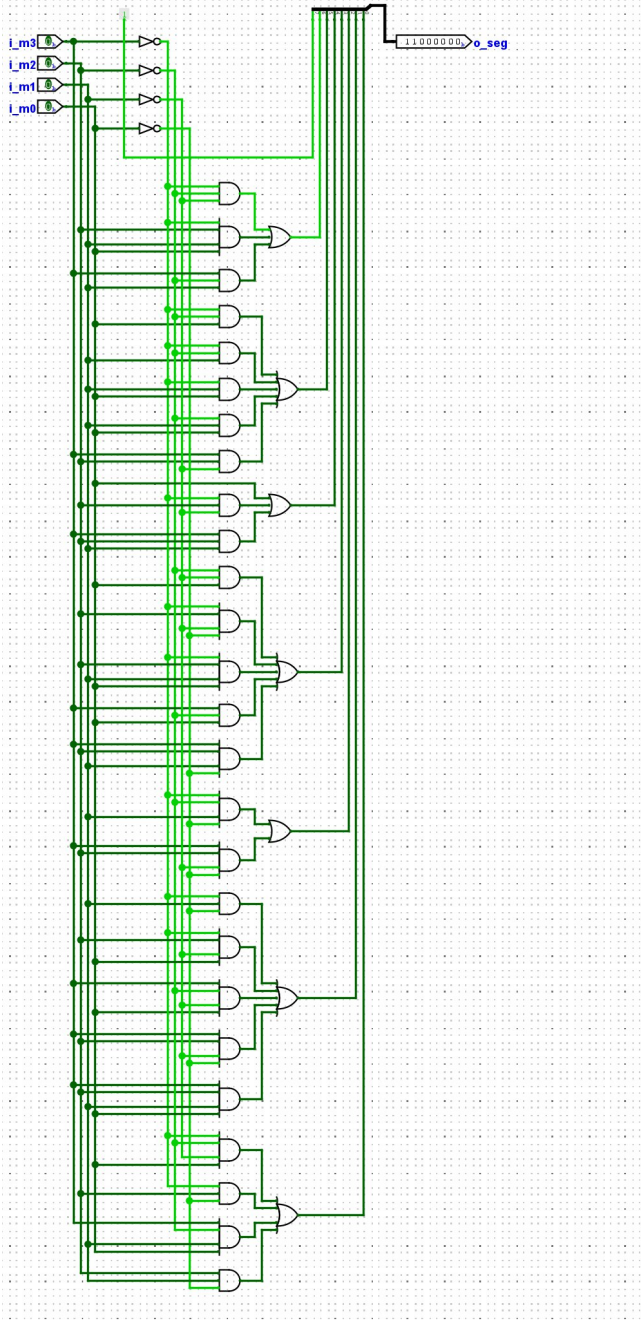
# Lab 4 Digital Design

Andrew Walker

# 7 Segment Decoder

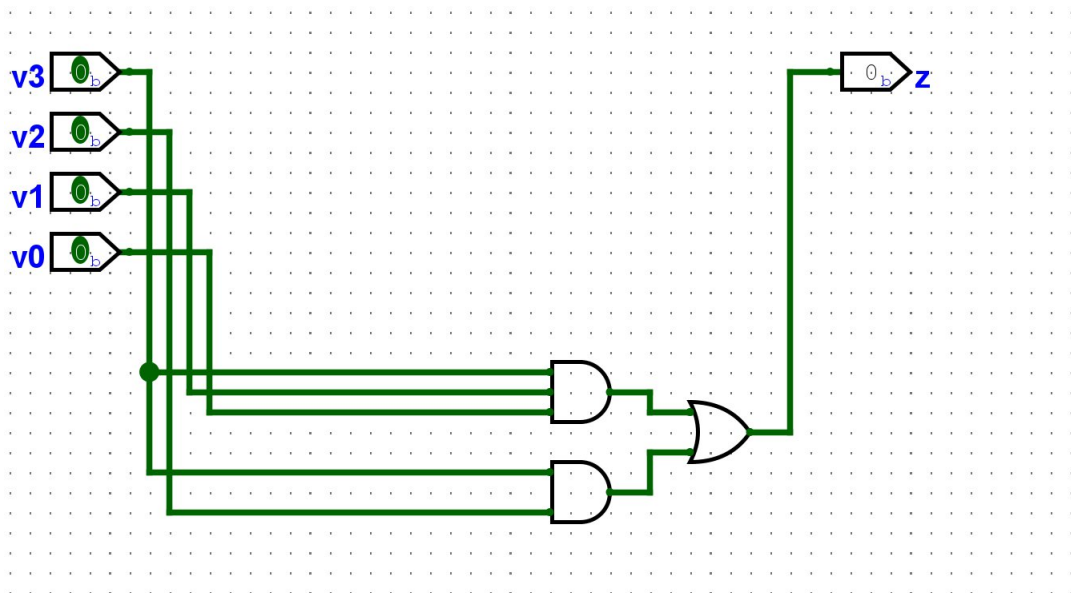


Created the output based on the the figure above. The segments are numbered and the outputs correspond to numbers 1-9.



| i_m3 | i_m2 | i_m1 | i_m0 | o_seg[7..0] |
|------|------|------|------|-------------|
| 0    | 0    | 0    | 0    | 11000000    |
| 0    | 0    | 0    | 1    | 11111001    |
| 0    | 0    | 1    | 0    | 10100110    |
| 0    | 0    | 1    | 1    | 10110000    |
| 0    | 1    | 0    | 0    | 10011001    |
| 0    | 1    | 0    | 1    | 10010010    |
| 0    | 1    | 1    | 0    | 10000011    |
| 0    | 1    | 1    | 1    | 11111000    |
| 1    | 0    | 0    | 0    | 10000000    |
| 1    | 0    | 0    | 1    | 10011010    |
| 1    | 0    | 1    | 0    | 11000000    |
| 1    | 0    | 1    | 1    | 11111001    |
| 1    | 1    | 0    | 0    | 10100110    |
| 1    | 1    | 0    | 1    | 10110000    |
| 1    | 1    | 1    | 0    | 10011001    |
| 1    | 1    | 1    | 1    | 10010010    |

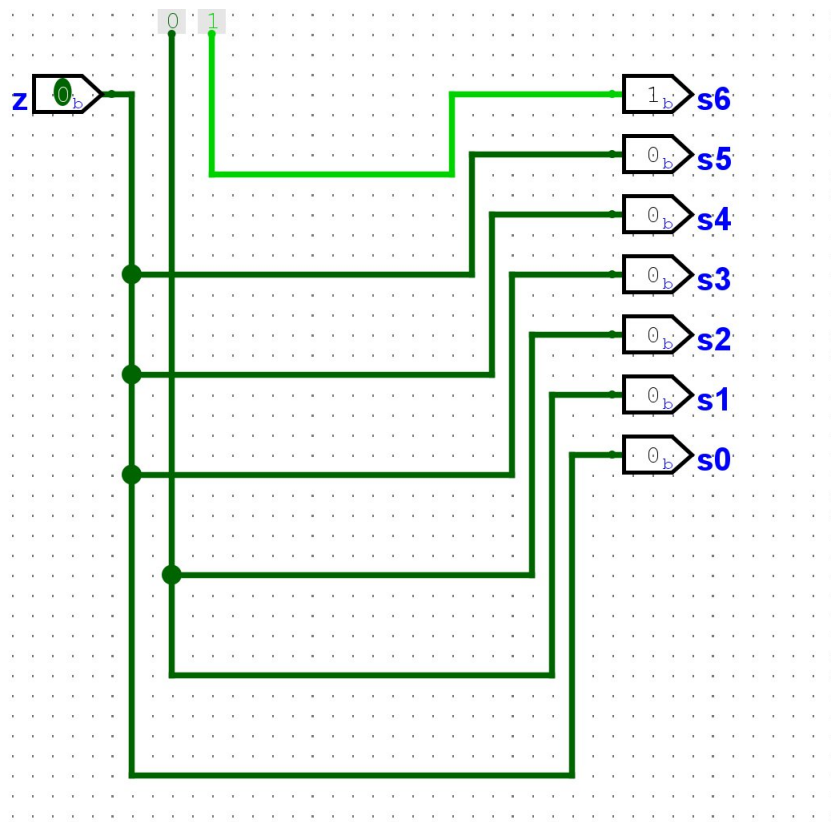
Circuit B



Looked at each row where  $z=1$  and looked if its complement equaled zero.

| v3 | v2 | v1 | v0 | z |
|----|----|----|----|---|
| 0  | 0  | 0  | 0  | 0 |
| 0  | 0  | 0  | 1  | 0 |
| 0  | 0  | 1  | 0  | 0 |
| 0  | 0  | 1  | 1  | 0 |
| 0  | 1  | 0  | 0  | 0 |
| 0  | 1  | 0  | 1  | 0 |
| 0  | 1  | 1  | 0  | 0 |
| 0  | 1  | 1  | 1  | 0 |
| 1  | 0  | 0  | 0  | 0 |
| 1  | 0  | 0  | 1  | 0 |
| 1  | 0  | 1  | 0  | 0 |
| 1  | 0  | 1  | 1  | 1 |
| 1  | 1  | 0  | 0  | 1 |
| 1  | 1  | 0  | 1  | 1 |
| 1  | 1  | 1  | 0  | 1 |
| 1  | 1  | 1  | 1  | 1 |

# Comparator



| z | s6 | s5 | s4 | s3 | s2 | s1 | s0 |
|---|----|----|----|----|----|----|----|
| 0 | 1  | 0  | 0  | 0  | 0  | 0  | 0  |
| 1 | 1  | 1  | 1  | 1  | 0  | 0  | 1  |

Using the z output from the last circuit I determined these values in the truth table.

# Video of Seven Segment Display Working

